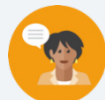


8.3 Special Parallelograms

Lesson Introduction

 Benchmarks	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.		 Learning Targets	- I can prove rectangles are parallelograms. - I can use properties of parallelograms to identify special parallelograms.
	MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.			
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What properties determine that a parallelogram is a rectangle? - What properties determine that a parallelogram is a rhombus? - What properties determine that a parallelogram is a square?			
 Lesson Narrative	This lesson builds on the properties of parallelograms for students to explore the differences between rectangles, rhombi, and squares.			
 Component Summary	8.3.1 Warm-Up (~5 min.)	Compare an estimate to a calculated circumference. (SE p. 19)	8.3.4 Guided Instruction (10-15 min.)	Determine the properties of a square and a rhombus. (SE p. 22)
	8.3.2 Collaboration (10-15 min.)	Determine the properties of a rectangle. (SE p. 20)	8.3.5 Bring It Together (5 min.)	Explain how to determine if a parallelogram is a rectangle, square, or rhombus. (SE p. 24)
	8.3.3 Exploration (5-10 min.)	Compare properties of parallelograms and rectangles. (SE p. 21)	8.3.6 Wrap-Up (~5 min.)	Reflect on the interest and understanding of the lesson. (SE p. 25)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Rectangle - Square - Rhombus <u>Additional Glossary:</u> N/A		- Rulers (Optional) String or Shoelace (Optional) Chart paper	
			Additional Preparation	
			Make handouts of a blank chart that can be filled in to categorize polygons.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse rectangles and squares. Drawing pictures with examples and non-examples can help students visualize the differences and deepening their understanding. - Students may place statements in more than one area of the Venn diagram in 8.3.4 Guided Instruction.	Consider having students create a foldable or use chart paper to display properties of each type of parallelogram discussed in the lesson, along with pictorial representations that are examples and non-examples.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time During 8.3.5 Bring It Together, consider using the “Stronger and Clearer Each Time” routine to provide a structured and interactive opportunity for students to revise and refine both their ideas (verbal and written). This routine provides a purpose for student conversation, as well as fortifies the output. Throughout this process, students should be pressed for details, and encouraged to press each other for details.	MA.K12.MTR.4.1: Discussion and Reflection In 8.3.2 Collaboration and 8.3.3 Exploration, students will have the opportunity to discuss and reflect as they determine the attributes of rectangles and discover that rectangles are a type of parallelogram. Students will make conjectures about the lengths of diagonals in a rectangle, and about what happens at the intersection of diagonals in a rectangle, and then apply their understanding on rectangle $RPQD$ in question 2 during the 8.3.3 Exploration.
 Differentiate Instruction	This lesson is focused around developing clear definitions for different types of parallelograms. There are multiple opportunities throughout the lesson for kinesthetic learners to have an entry point, for example, when measuring segment lengths in 8.3.4 Guided Instruction. Consider allowing students voice and choice in how they respond to the questions in 8.3.5 Bring It Together. This could include students explaining to a partner, drawing diagrams with labels, recording an audio message, or writing an explanation.	
Lesson Notes:		

8.3.1 Warm-Up: Estimation

Component Overview: Students estimate and then use their understanding of circumference and diameter to determine the number of cereal pieces it would take to cover the edge of their bowl.



8.3.1 Warm-Up: Estimation

1. A picture of a bowl with a cereal square is shown.



- a. Estimate the number of cereal squares that will go around the inner circumference of the bowl.

Answers will vary.
40

2. A picture of the bowl with a diameter cereal squares is shown.



- a. Use the diameter of cereal squares to find the circumference.

43 or 44 cereal squares



This activity could be easily replicated in the classroom with paper bowls or plates, and any small box of cereal.

8.3.1 Warm-Up: Estimation (continued)

- b. Compare your estimate to the circumference found using the diameter of cereal squares.

Answers will vary. The diameter of the circle was 14 cereal squares. When calculating the circumference using it, the total was 43.982 which is just a little more than the estimation of the diameter of the cereal squares that could go around the bowl.

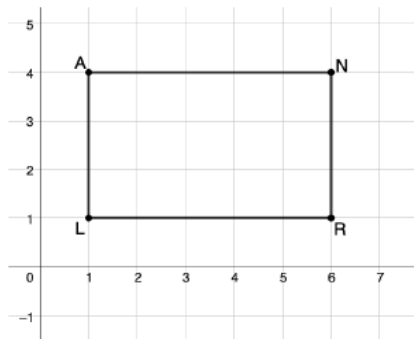
8.3.2 Collaboration: What Is a Rectangle?

Component Overview: Students will measure the lengths of various segments. Using these measurements, students will write a conjecture about the properties of rectangles. Students will need a partner for this component.



8.3.2 Collaboration: What is a Rectangle?

1. *ANRL* is a rectangle shown on the coordinate grid.



- a. List everything you know about this type of quadrilateral.
Answers will vary.
Two sets of parallel lines, opposite sides are congruent, and the sides intersect at right angles.
- b. Compare your list with your partner's list. Add anything you may have missed.
Answers will vary.
- c. With your partner, list two truths and one lie about this type of quadrilateral in the table below. Share your list with another pair. See if they can spot the lie.

Answers will vary.

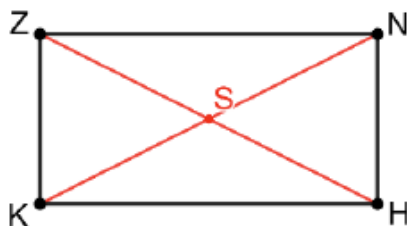
1. A rectangle has all sides congruent.
2. A rectangle has congruent angles.
3. A rectangle has two sets of congruent sides.



The rectangle shown is also a quadrilateral. Facilitate discussion on why a rectangle is a sub-class of a quadrilateral. Consider starting a chart that shows the classifications of polygons that is built upon through this lesson and future lessons.

8.3.2 Collaboration: What Is a Rectangle? (continued)

2. $ZNHK$ is shown.



- Using a ruler, draw $\overline{ZH}$ and $\overline{KN}$.
- These segments are called diagonals.
- Measure the length of these segments and write them in the table below. Include units.
Answers will vary depending on unit used, but they should be congruent.

Segment	Length
$\overline{ZH}$	7.8 cm
$\overline{KN}$	7.8 cm

- Make a conjecture about the lengths of the diagonals of a rectangle.
The lengths of the diagonals of a rectangle are congruent.
- Label the intersection point of the diagonals S .
Shown on the figure.
- Measure the following segments and fill in their length in the table below. Include units.
Answers will vary depending on unit used, but they should be congruent.

Segment	Length
$\overline{ZS}$	3.9 cm
$\overline{HS}$	3.9 cm
$\overline{KS}$	3.9 cm
$\overline{NS}$	3.9 cm



Consider challenging some students to complete this task with only a piece of string or a shoelace. Students could tie knots in the string and measure the number of “sections” or units that they count for the measurement. The intent here is not to have the same measurement as someone else, but to realize that each segment has the same length, regardless of the unit of measure.



For ELL students, consider offering a word bank for question 2b, to allow for an entry point.

8.3.2 Collaboration: What Is a Rectangle? (continued)

- g. What do you notice?

Answers will vary.

All the lengths from the point of intersection on the diagonals are the same.

- h. Make a conjecture about what happens to the diagonals of a rectangle when they intersect.

The diagonals of a rectangle bisect each other.



While monitoring students' work, listen for appropriate mathematical vocabulary, like "point of intersection" and "bisect/bisector."

8.3.3 Exploration: Comparing Parallelograms and Rectangles

Component Overview: Students discover that a rectangle is a type of parallelogram and complete statements for a given rectangle.



8.3.3 Exploration: Comparing Parallelograms and Rectangles

1. Work with your partner to fill in the table. If the statement describes a property always true of the shape, mark the box.

Statement	Parallelogram	Rectangle
opposite sides congruent	✓	✓
opposite sides parallel	✓	✓
opposite angles congruent	✓	✓
consecutive angles supplementary	✓	✓
diagonals bisect each other	✓	✓
diagonals are congruent		✓
all angles are congruent		✓

Based on the chart, discuss with your partner whether a rectangle could be classified as a parallelogram.

Answers will vary.

Yes, a rectangle can be classified as a parallelogram because all the properties of a parallelogram are also properties of a rectangle.



A **rectangle** is a quadrilateral that contains four right angles. Rectangles may be oblong or square. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms. Squares form a sub-class.



This Exploration is scaffolded to allow students to engage with their partner, rather than being guided through by the teacher.

Observe and listen as students work with their partners. Make note of students' ideas that you can ask them to share before moving to the next component.

8.3.3 Exploration: Comparing Parallelograms and Rectangles (continued)

2. $RPQD$ is a rectangle.



Complete the statements.

It is given that $RPQD$ is a rectangle. By the definition of

a rectangle $\angle R, \angle P, \angle Q$, and $\angle D$ are all

- ☐ right angles,
- ☐ acute angles,
- ☐ obtuse angles,

which means that $m\angle R = m\angle P = m\angle Q = m\angle D =$

- ☐ 90° .
- ☐ 180° .

Since $m\angle P + m\angle Q = 180^\circ$ we can show that

- ☐ $\overline{RD} \parallel \overline{PQ}$
- ☐ $\overline{RP} \parallel \overline{DQ}$

and since $m\angle D + m\angle Q = 180^\circ$ we can show that

- ☐ $\overline{RD} \parallel \overline{PQ}$
- ☐ $\overline{RP} \parallel \overline{DQ}$

by the converse of the

- ☐ alternate interior angles
- ☐ consecutive interior angles

theorem. Therefore, $RPQD$ is a

- ☐ parallelogram
- ☐ kite

by

definition.



After students answer question 2 prior to moving to the next component, clarify any questions or misunderstandings students have about rectangles, before moving to the next component.

8.3.4 Guided Instruction: More Special Parallelograms

Component Overview: Students will measure various segments and angles to decide what properties apply to squares and to rhombi.



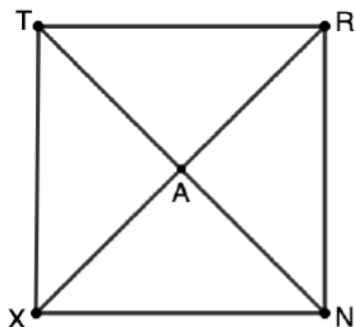
8.3.4 Guided Instruction: More Special Parallelograms



A **square** is a quadrilateral with four right angles and four equal-length sides. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms, rectangles, rhombi.

1. $TRNX$ is a square with diagonals $\overline{TN}$ and $\overline{XR}$ intersecting at point A .

- a. Use a ruler and a protractor to fill in the chart with the appropriate measurements. Include the units.
Lengths depend on student lesson format.



Segment	Measure	Segment/Angle	Measure
$\overline{TN}$	6.6 cm	$\overline{XA}$	3.3 cm
$\overline{XR}$	6.6 cm	$\overline{RA}$	3.3 cm
$\overline{TA}$	3.3 cm	$\angle XAT$	90°
$\overline{NA}$	3.3 cm	$\angle NAR$	90°

- b. Which of the following describe the diagonals of a square? Circle all that apply.

congruent

bisect

perpendicular



If you have limited rulers, allow students to work in pairs to measure the segments, but ask that students take turns measuring.

8.3.4 Guided Instruction: More Special Parallelograms (continued)

- c. List the features that determine if a parallelogram is a square.

All angles are congruent
Diagonals are congruent
Diagonals are perpendicular
All sides are congruent



Consider having students chart the common features between a square and a parallelogram on chart paper.

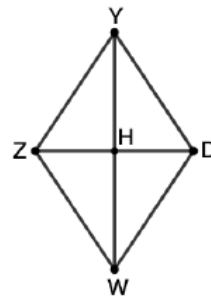


A **rhombus** is a quadrilateral containing four equal-length sides. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms. Squares form a sub-class.

2. $YDWZ$ is a rhombus with diagonals YW and ZD intersecting at point H .

- a. Use a ruler and a protractor to fill in the chart with the appropriate measurements. Include units.

Lengths depend on student lesson format.



Segment	Measure	Segment/Angle	Measure
$\overline{YW}$	3.6 cm	$\overline{ZH}$	1.2 cm
$\overline{ZD}$	2.4 cm	$\overline{DH}$	1.2 cm
$\overline{YH}$	1.85 cm	$\angle ZHW$	90°
$\overline{WH}$	1.85 cm	$\angle YHD$	90°

- b. Which of the following describe the diagonals of a rhombus? Select all that apply.

congruent

bisect

perpendicular

8.3.4 Guided Instruction: More Special Parallelograms (continued)

- c. List the features that determine if a parallelogram is a rhombus.

Diagonals are perpendicular; All sides are congruent

- d. What are the differences between a square and a rhombus?

In a square, diagonals are congruent. In a rhombus, diagonals may or may not be congruent. A square has all angles equal to 90° . A rhombus has opposite angles equal, though they could be equal to 90° .

3. Use the numbered statements in the box to fill in the Venn diagram comparing rectangles, squares, and rhombi.

1. opposite sides are congruent

2. opposite sides are parallel

3. all angles are congruent

4. diagonals are perpendicular

5. opposite angles are congruent

6. consecutive angles are supplementary

7. all sides are congruent

8. diagonals are congruent

9. diagonals bisect each other

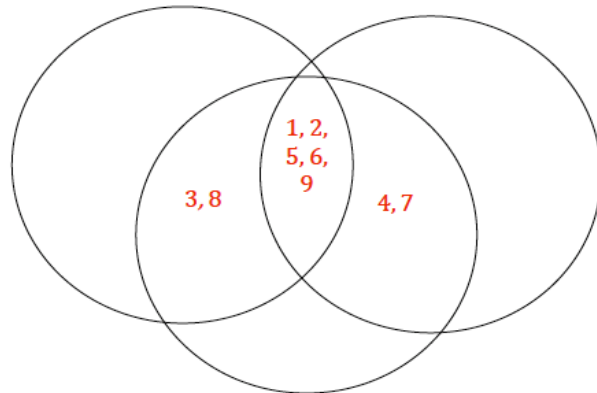


Have students add the common features and differences from question 2 parts c and d to their chart paper. This will assist them later in the lesson when they get to the Venn diagram.

8.3.4 Guided Instruction: More Special Parallelograms (continued)

Rectangle

Rhombus

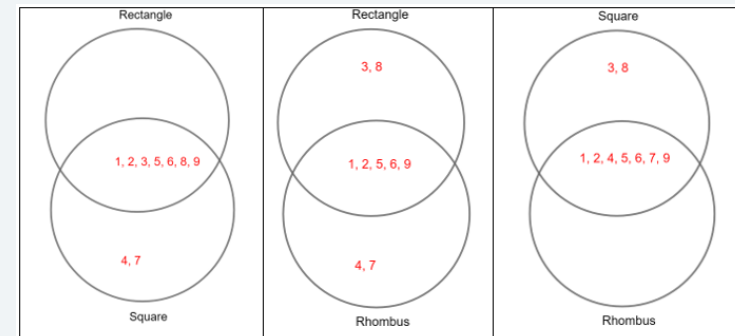


Square



Students may place statements in more than one area of the Venn diagram. For students who struggle with looking at all three figures at one time with the given statements, students can begin by creating a Venn diagram with only two figures:

- Rectangle and Rhombus
- Rectangle and Square
- Rhombus and Square



8.3.5 Bring It Together: Which Special Parallelogram Is It?

Component Overview: Students write statements that classify a figure as either a rectangle, rhombus, or square.



8.3.5 Bring It Together: Which Special Parallelogram Is It?

1. Explain how to show a parallelogram is a rectangle but not a rhombus or a square.
Answers will vary. A parallelogram is a rectangle but not a square or rhombus when it does not have four congruent sides.
2. Explain how to show a parallelogram is a rhombus but not a square or a rectangle.
Answers will vary. A parallelogram is a rhombus but not a square or rectangle when the diagonals are not congruent.
3. Explain how to show a parallelogram is a square.
Answers will vary. A parallelogram is square when diagonals are congruent and all sides are congruent.



Stronger and Clearer Each Time

Consider using this routine as students write explanations in questions 1-3. Have students think and write individually first. Then, through a structured pairing strategy, present the students multiple opportunities to refine and clarify their response through conversation. Finally, students revise their original written response and prepare to share the response with the whole group.

8.3.6 Wrap-Up: Reflection

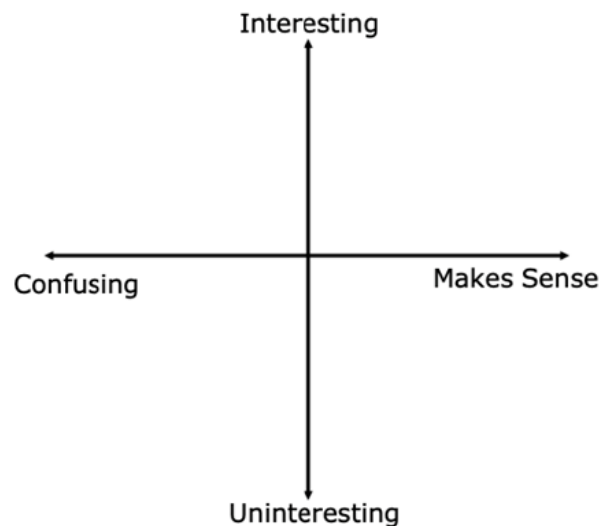
Component Overview: Students will place a mark on the grid that summarizes their comfort level and interest in the learning target from the lesson, and explain why they placed a point at that location.



8.3.6 Wrap-Up: Reflection

1. Plot a point on the grid to indicate your level of interest and how much sense you made of today's learning targets.

Answers will vary.



2. Explain why you placed the point where you did.

Answers will vary.



Monitor for students who mark that they were confused. Make note of those students' explanations to create a plan for addressing their confusions during the next lesson.